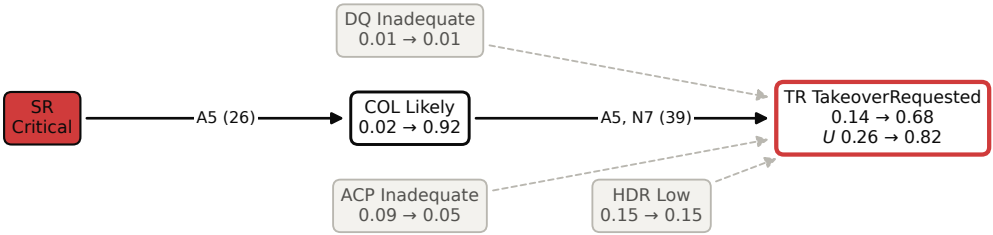
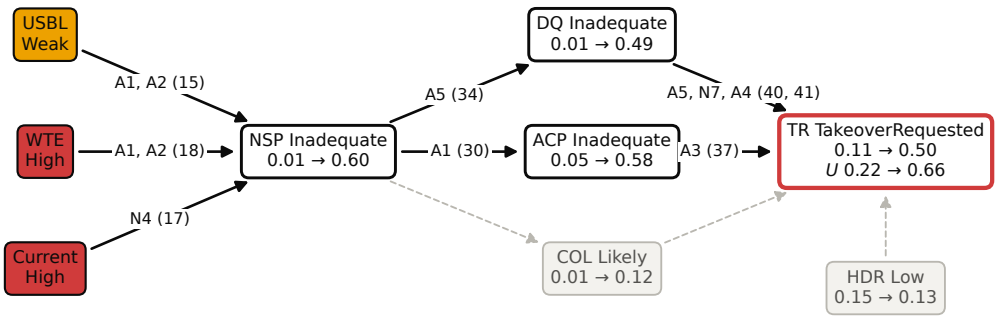
(a) S1: obstacle on the transit leg



(26) SR → COL: return in the safety box (C6, C8) on the path to collision (C3); transit task AT3
(39) COL → TR: collision pathway to the takeover constraint C1

(b) S2: cross-current and loss of the USBL signal during inspection



(15) USBL → NSP: position fix AI1.1 feeds the navigation decision AI1.2 (C5)
(18) WTE → NSP: tracking error read in AI1.1 for the on-track decision AI1.2 (I-2)
(17) Current → NSP: read with the vehicle state (AI2.1); degrades position keeping
(30) NSP → ACP: estimation precedes execution (AU2.3, AT1.3, AH1.3, AD3.3 publish the state used by AU3, AT2, AH2, AD5)
(34) NSP → DQ: valid position required for adequate data (I-2, I-4 → I-6 → I-X3)
(40, 41) DQ → TR: data-quality pathway to C1, active in inspection only
(37) ACP → TR: request-and-confirm interaction A-TO.3 → H-TO.1